

Heat Load & Commissioning

CSA F280-12 design summary and final commissioning record for a whole-home central ducted heat-pump conversion.

PROJECT

PARTICIPANT	Jordan Lee
ADDRESS	27 Cedar Ridge Road, Port Hardy, BC V0N 2P0
HOME	Single detached, 920 ft2 conditioned area
DESIGN METHOD	CSA F280-12 room-by-room heat loss

PRIOR PRIMARY HEAT

The existing natural-gas forced-air furnace was the home's primary heating system. It served the complete central duct system, all conditioned living areas and more than 50% of the home throughout the heating season to at least 21 C.

Replacement: Bryant central ducted heat pump with electric auxiliary backup.

12,820

CALCULATED DESIGN HEAT LOSS BTU/H

13,600

HEAT-PUMP CAPACITY AT -5 C BTU/H

106%

CAPACITY / DESIGN LOAD

ROOM-BY-ROOM DESIGN SUMMARY

ZONE	CONDITIONED AREA	DESIGN HEAT LOSS	DISTRIBUTION
Living / dining / kitchen	390 ft2	5,160 BTU/h	Existing central supply and return ductwork
Bedrooms	310 ft2	4,280 BTU/h	Existing central supply branches
Bath / hall / utility	220 ft2	3,380 BTU/h	Existing central supply branches
Total conditioned space	920 ft2	12,820 BTU/h	100% of conditioned living area served

SIZING CONCLUSION

The selected AHRI 203380999 system has sufficient cold-weather capacity for the calculated design load and is sized to function as the home's primary heating system. The existing duct network provides heat to all conditioned space previously served by the natural-gas furnace.

FINAL COMMISSIONING - JULY 29, 2026

CHECK	OBSERVED RESULT	STATUS
Outdoor / indoor model match	38MAQB18R--3 / FMC4Z1800AL; AHRI 203380999	Verified
Airflow and static pressure	Measured external static 0.47 in. w.c.; airflow balanced across all conditioned zones	Pass
Primary heat operation	Heat-pump heating sequence and thermostat staging verified	Pass
Backup heat	Electric auxiliary heat energized and tested; no fossil-fuel backup connected	Pass
Fossil-fuel removal	Former natural-gas furnace absent; permit PH-MECH-26-0714 final accepted	Verified

Priya Nair, RASDT
Residential Air System Design Technician

July 29, 2026

Designer / commissioning technician

Commissioning completion